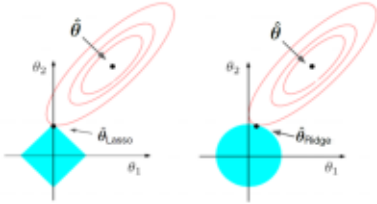
LASSO VS. RIDGE GEOMETRY

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left(y^{(i)} - f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)^2 \qquad \text{s.t. } \|\boldsymbol{\theta}\|_p^p \leq t$$

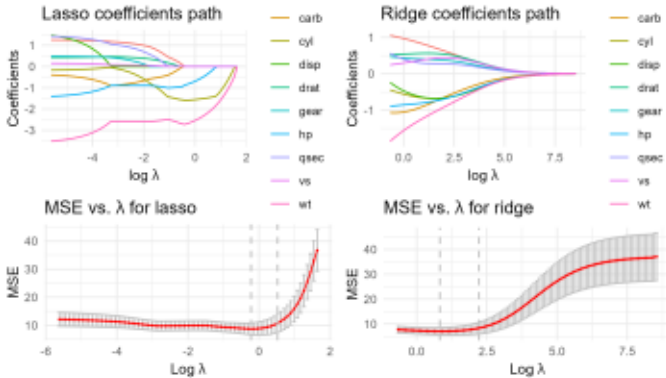


- In both cases (and for sufficiently large λ), the solution which minimizes $\mathcal{R}_{\text{reg}}(\boldsymbol{\theta})$ is always a point on the boundary of the feasible region.
- As expected, $\hat{\boldsymbol{\theta}}_{\text{lasso}}$ and $\hat{\boldsymbol{\theta}}_{\text{ridge}}$ have smaller parameter norms than $\hat{\boldsymbol{\theta}}$.
- For lasso, solution likely touches a vertex of constraint region. Induces sparsity and is a form of variable selection.
- For $p > n$: lasso selects at most n features ▸ Zou and Hastie 2005.

COEFFICIENT PATHS AND 0-SHRINKAGE

Example 1: Motor Trend Car Roads Test (mtcars)

We see how only lasso shrinks to exactly 0.



NB: No real overfitting here, as data is so low-dim.

REGULARIZATION AND FEATURE SCALING

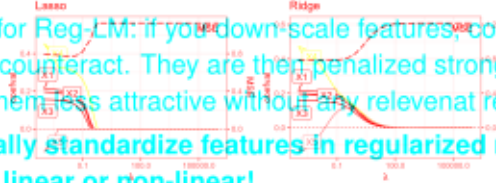
- Typically we omit θ_0 in penalty $J(\theta)$ so that the "infinitely" regularized model is the constant model (but can be implementation-dependent).
- Unregularized LM has **rescaling equivariance**, if you scale some features, can simply "anti-scale" coefs and risk does not change.
- Not true for Reg-LM: if you down-scale features, coeffs become larger to counteract. They are then penalized stronger in $J(\theta)$, making them less attractive without any relevenat reason.
- **So: usually standardize features in regularized models, whether linear or non-linear!**

CORRELATED FEATURES: L1 VS L2

Simulation with $n = 100$:

$$y = 0.2x_1 + 0.2x_2 + 0.2x_3 + 0.2x_4 + 0.2x_5 + \epsilon$$

x_1-x_4 are independent, but x_4 and x_5 are strongly correlated.



- L1 removes x_5 early, L2 has similar coeffs for x_4, x_5 for larger λ
- Also called "grouping property": for ridge highly corr. features tend to have equal effects; lasso however "decides" what to select
- L1 selection is somewhat "arbitrary"

SUMMARY

- Neither ridge nor lasso can be classified as better overall
- Lasso can shrink some coeffs to zero, so selects features; ridge usually leads to dense solutions, with smaller coeffs
- Lasso likely better if true underlying structure is sparse ridge works well if there are many (weakly) influential features
- Lasso has difficulties handling correlated predictors; for high correlation, ridge dominates lasso in performance
- Lasso: for (highly) correlated predictors, usually an "arbitrary" one is selected, with large coeff, while the others are (nearly) zeroed
- Ridge: coeffs of correlated features are similar

CORRELATED FEATURES: $L1$ VS $L2$

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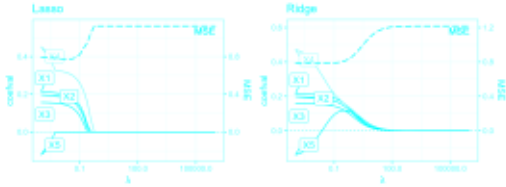

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More detailed answer: The "random" decision is in fact a complex deterministic interaction of data geometry (e.g., corr. structures), the optimization method, and its hyperparamters (e.g., initialization). The theoretical reason for this behavior relates to the convexity of the penalties ▸ Zou and Hastie 2005.

Considering perfectly collinear features $x_4 = x_5$ in the last example, we can obtain some more formal intuition for this phenomenon:

- Because $L2$ penalty is *strictly* convex:
  $x_4 = x_5 \implies \hat{\theta}_{4,ridge} = \hat{\theta}_{5,ridge}$ (grouping prop.)
- $L1$ penalty is not *strictly* convex. Hence, no unique solution exists if $x_4 = x_5$, and sum of coefficients can be arbitrarily allocated to both features while remaining minimizers (no grouping property!): For any solution $\hat{\theta}_{4,lasso}, \hat{\theta}_{5,lasso}$, equivalent minimizers are given by
  $\tilde{\theta}_{4,lasso} = s \cdot (\hat{\theta}_{4,lasso} + \hat{\theta}_{5,lasso})$ and $\tilde{\theta}_{5,lasso} = (1-s) \cdot (\hat{\theta}_{4,lasso} + \hat{\theta}_{5,lasso}) \; \forall s \in [0,1]$

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Tibshirani 1996 | Zou and Hastie 2005

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